

Q1.

Fortune High School gave its students a wider choice of subjects to study. The table shows the number of students, of each gender, who chose to study each of the additional subjects during the school year 2007/08.

	Bulgarian	Climate Change	Finance	Polish
Male	7	31	25	40
Female	2	24	22	19

Assuming that these data form a random sample, use a χ^2 test, at the 10% level of significance, to test whether the choice of these subjects is independent of gender.

(Total 11 marks)

Q2.

A sample survey, conducted to determine the attitudes of residents to a proposed reorganisation of local schools, gave the following results.

		Against reorganisation	Not against reorganisation
Age of resident	16–17	9	2
	18–21	17	10
	22–49	115	90
	50–65	41	34
	Over 65	3	4

Use a χ^2 test, at the 5% level of significance, to determine whether there is an association between the ages of residents and their attitudes to the proposed reorganisation of local schools.

(Total 12 marks)

Q3.

Year 12 students at Newstatus School choose to participate in one of four sports during the Spring term.

The students' choices are summarised in the table.

	Squash	Badminton	Archery	Hockey	Total
Male	5	16	30	19	70

Female	4	20	33	53	110
Total	9	36	63	72	180

- (a) Use a χ^2 test, at the 5% level of significance, to determine whether the choice of sport is independent of gender.

(10)

- (b) Interpret your result in part (a) as it relates to students choosing hockey.

(2)

(Total 12 marks)

Q4.

A political party held a conference, attended by delegates from local branches, to choose a new leader. The voting at the conference was carried out in public. In a confidential survey, randomly selected delegates were asked whether they would have voted differently if voting had been by secret ballot. The answers, classified by the ages of the delegates, are summarised in the following table.

	Definitely	Possibly	Definitely not
Under 65 years	4	12	6
65 years and over	10	28	54

- (a) Use a χ^2 distribution and the 5% significance level to examine whether the answer to the question is associated with the age of the delegate.

(10)

- (b) Interpret your result in part (a) in the context of the question.

(1)

- (c) Make a further comment about the ages of the delegates.

(1)

(Total 12 marks)

Q5.

The board of an insurance company was concerned about the poor performance of its motor insurance business. Belinda was asked to collect and analyse data on the claims record of the 4902 customers who held motor policies in 2003.

- (a) The following contingency table summarises the number of claims made in 2003 classified by length of time the policy had been held.

		Number of Claims in 2003			
		0	1	2	3 or more
Policy	less than 2 years	1478	212	21	6

held for	2 or more years	2830	334	20	1
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Use a χ^2 distribution and the 1% significance level to investigate whether the number of claims made in 2003 is associated with the length of time for which the policy had been held.

(11)

- (b) Belinda also constructed the following tables concerning the 4902 customers.

Table A

		% of Total Number of Policy Holders who in 2003	
		Made a Claim	Did not make a Claim
Gender	Male	8.4	57.6
	Female	3.7	30.3

Table B

		Total Number of Policy Holders in 2003	Number of Policy Holders who made a claim in 2003
Age (years)	< 25	956	280
	25 or more	3946	314

Table C

		Number of Policy Holders who in 2003	
		Did not make a Claim	Made a claim for more than £2000
Age (years)	< 25	676	145
	25 or more	3632	192

For **each** table:

- (i) state why, in its present form, it cannot be analysed as a contingency table using a χ^2 distribution;
- (ii) construct a new table which can be analysed as a contingency table using a χ^2 distribution. **Do not analyse any of the tables you construct.**

(3)

(6)

(Total 20 marks)